Tox21 Phase 1: Reproducible Molecular Toxicity Prediction with Fingerprints and Classical Machine Learning

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Abstract—We build and evaluate a reproducible classical machine-learning baseline for the 12 endpoint Tox21 molecular toxicity benchmark. The source table contained 7,831 records; RDKit cleaning, largest-fragment selection, canonicalization, duplicate reconciliation, and label-conflict masking yielded 7,709 unique structures. Molecules are represented by 2,048-bit radius-2 Morgan fingerprints, optionally concatenated with 213 train-preprocessed RDKit descriptors. Logistic Regression (LR), Random Forest (RF), and XGBoost (XGB) are trained endpoint-wise with missing labels excluded and class imbalance handled by model-specific weights. Eight configurations per model and feature set are selected by scaffold-validation average precision, then evaluated on fixed scaffold and random splits over three seeds. Across endpoint-macro means, RF with fingerprints plus descriptors is strongest on the scaffold split (ROC AUC 0.776 ± 0.002; PR AUC 0.404 ± 0.004); XGB with the hybrid representation is close in PR AUC (0.401 ± 0.005). Random-split scores are higher for every model-feature pair, with the mean PR-AUC gap ranging from 0.134 to 0.187. These results support descriptors as a useful complement for tree models while confirming that evaluation on unseen scaffolds is materially harder. The benchmark establishes a transparent Phase 1 reference for future graph-based models.

Index Terms— Tox21, molecular toxicity, Morgan fingerprints, RDKit descriptors, scaffold split, class imbalance, average precision.

I. INTRODUCTION

Chemical safety screening must prioritize a large and chemically diverse universe of substances while experimental assays remain costly and incomplete. The Tox21 collaboration established a high-throughput benchmark with 12 nuclear-receptor and stress-response assays. A molecule may be untested on an assay, so a blank label means unknown rather than inactive. Active compounds are uncommon: in this cleaned set, endpoint prevalence ranges from 2.9% to 15.9%. These properties make label masking, careful split design, and metrics suited to rare positives central to the modeling task [1]–[3].

This Phase 1 study builds an auditable machine-learning baseline from two-dimensional molecular structures. It compares local substructure fingerprints with a hybrid representation that adds global physicochemical descriptors, across three classifier families and two evaluation splits. The random split provides a conventional reference; the Bemis–Murcko scaffold split asks the more demanding question of performance on held-out chemical frameworks [4].

The main contributions are: (i) a cleaned, deduplicated set with endpoint-specific missing labels preserved; (ii) a fixed train-only descriptor preprocessing path and shared evaluation harness; (iii) a matched comparison of LR, RF, and XGB under a common eight-configuration tuning budget; and (iv) an analysis of feature gains and the gap between random and scaffold evaluation. Phase 2 graph neural networks are outside the scope of these measurements.

II. RELATED WORK

A. Toxicity benchmarks and labels

The Tox21 challenge evaluates chemical activity across 12 biological assays. DeepTox showed that predictive pipelines combining molecular representations with machine learning can be effective for toxicity screening [1], [2]. MoleculeNet later standardized molecular machine-learning benchmarks and emphasized that data partition and metric choices influence conclusions [3]. In sparse screening matrices, a model must learn and be scored separately on observed labels for each endpoint; treating untested compounds as negatives introduces label noise.

B. Molecular representations

Morgan fingerprints encode circular atom neighborhoods through iterative hashing; radius 2 corresponds to environments extending to four bonds, commonly called ECFP4 [5], [6]. Their fixed-length bit vectors preserve local motif presence but compress a much larger substructure space, allowing collisions. RDKit descriptors provide a complementary view through continuous molecular properties such as mass, lipophilicity, polar surface area, and topology. Whether global descriptors improve a fingerprint model is an empirical question addressed here.

C. Evaluation under imbalance and chemical shift

ROC AUC summarizes positive-versus-negative ranking and has a 0.5 random baseline. Average precision (AP), reported as PR AUC in the project results, emphasizes precision among retrieved actives; its random baseline is approximately the active prevalence. Precision–recall analysis is particularly informative when positives are rare [7], [8]. Random row splits may distribute related analogues across partitions. Scaffold splitting groups Bemis–Murcko frameworks, reducing this overlap and testing a different, harder generalization regime [3], [4].

D. Classical and learned models

RF aggregates randomized trees, reducing variance through bootstrap sampling and feature subsampling [9]. XGB adds boosted trees sequentially while regularizing the objective [10]. Message-passing neural networks learn atom representations by repeatedly aggregating bonded-neighbor messages and applying a molecule-level readout [11]. Their published quantum-chemistry results motivate learned representations, but do not establish an advantage on Tox21; that comparison belongs to Phase 2.

III. DATASET AND EXPLORATORY ANALYSIS

The repository's source CSV contains 7,831 rows and 14 columns (molecule identifier, SMILES, and 12 endpoint labels). RDKit parsed 7,823 structures; eight invalid SMILES were removed. The pipeline selected the largest connected fragment, canonicalized SMILES, and merged duplicates endpoint by endpoint. Of 114 duplicate rows removed, 44 conflicting molecule–endpoint labels were changed to missing. The final dataset contains 7,709 unique structures. Missing labels remain NaN throughout model fitting and evaluation.

Each endpoint has different assay coverage. Observed positive prevalence spans 2.9% (NR-PPAR-gamma) to 15.9% (SR-ARE), making accuracy an inadequate summary. We therefore report endpoint-wise ROC AUC and AP on labeled rows, then give an unweighted endpoint macro-average. Figure 1 shows known-label coverage and the distribution of actives and inactives.

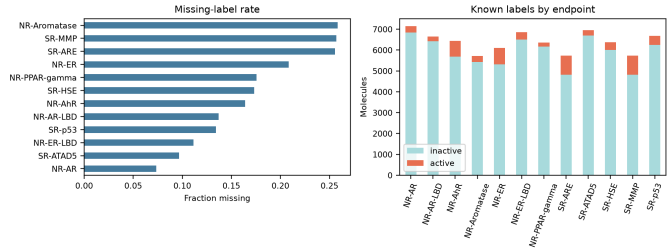


Fig. 1. Endpoint label coverage and active/inactive counts in the cleaned dataset.

Molecular properties are heterogeneous. The supplied exploratory figures summarize molecular weight, heavy-atom count, ring count, and scaffold families. The saved scaffold and random splits each contain 6,167 training, 771 validation, and 771 test molecules. The scaffold split assigns entire Bemis–Murcko groups to a single partition. All 1,753 acyclic molecules share the empty scaffold and are in training; validation and test consequently contain only ring-containing molecules. This is an important boundary on the interpretation of scaffold performance.

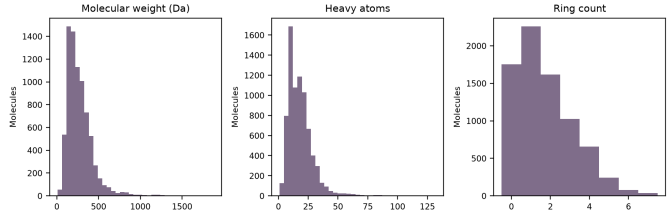


Fig. 2. Molecular property distributions after structure cleaning.

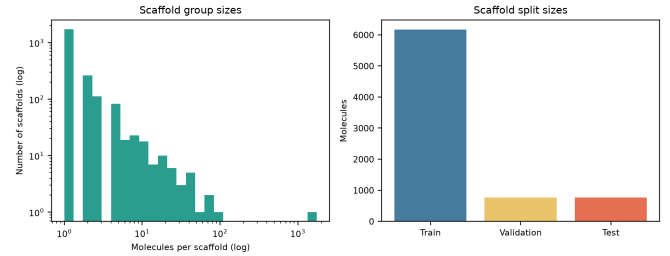


Fig. 3. Scaffold-group sizes and partition allocation; the empty-scaffold group remains in training.

IV. FEATURE ENGINEERING

A. Morgan fingerprints

RDKit's Morgan generator produces one binary vector per canonical SMILES with radius r = 2 and 2,048 bits, an ECFP4-equivalent setting [5], [6]. The vector records hashed local environments rather than exact global arrangement. The repository feature analysis reports 29.75 ± 16.41 active bits per molecule (median 26; range 1–160). Every bit position is used somewhere in the corpus, and the most frequent bit appears in 71.73% of compounds. Figure 4 illustrates fingerprint sparsity and bit-frequency concentration.

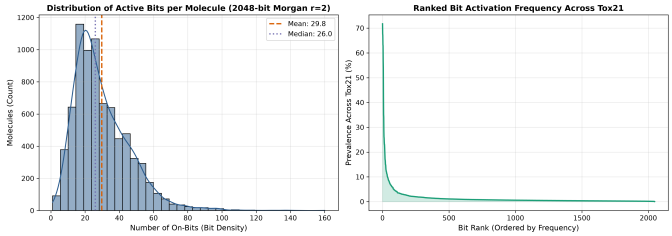


Fig. 4. Active-bit density per molecule and ranked Morgan-bit frequencies.

B. RDKit descriptors and preprocessing

The pipeline computes 217 raw two-dimensional RDKit descriptors. The feature audit found 1,224 NaN cells across 107 molecules and six infinite cells across two molecules. Three descriptors are constant zero across the data (SMR_VSA8, SlogP_VSA9, and fr_diazo). These values are preserved in the raw artifact; cleaning is applied only in the model pipeline.

For the hybrid feature set, descriptor preprocessing is fitted on the training partition only: convert infinities to missing values; drop all-missing columns; impute with training medians; clip to training 1st and 99.9th percentiles; remove columns with training standard deviation at or below 10^-8; and standardize using training means and standard deviations. Four low-variance columns are removed on the scaffold training partition, including fr_azide, leaving 213 descriptors. Validation and test rows use the frozen train-derived transforms. Concatenation gives 2,261 features. Figure 5 illustrates raw-versus-transformed descriptor distributions; Figure 6 shows descriptor correlation.

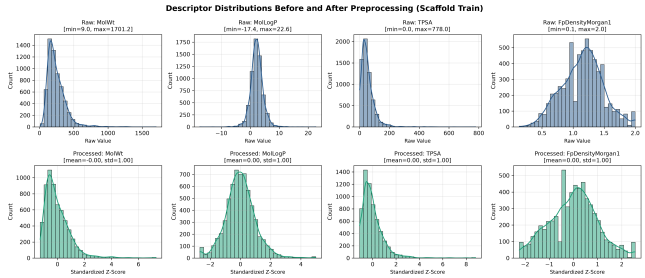


Fig. 5. Example descriptor distributions before and after train-fitted preprocessing.

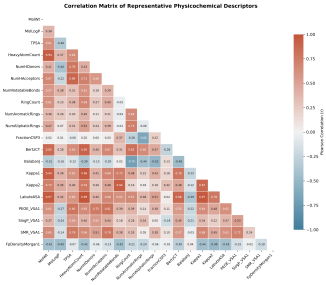


Fig. 6. Correlation structure among descriptors; correlated descriptors can share predictive signal.

V. MODELS AND EXPERIMENTAL SETUP

A. Endpoint-wise classifiers

For each endpoint, models train only on that endpoint's labeled training rows. LR uses scikit-learn's liblinear solver, balanced class weights, 2,000 maximum iterations, and either L1 or L2 regularization. RF uses independent binary RandomForestClassifier estimators with balanced-subsample weights, square-root feature subsampling, and seeded trees. XGB uses histogram tree construction, logistic classification, row subsampling 0.8, column subsampling 0.5, and scale_pos_weight = n_neg/n_pos computed from each endpoint's observed training labels. No synthetic oversampling is applied. LR estimates $p(y=1|x)=\sigma(w^Tx+b)$ under a class-weighted logistic loss with L1 or L2 regularization. RF averages tree probabilities, while XGB adds trees sequentially to reduce logistic loss under shrinkage and tree regularization. These inductive biases motivate comparing a linear model, bagged trees, and boosting.

B. Tuning and held-out evaluation

Each model-feature pair has eight candidate configurations. All are trained on the scaffold training partition and selected by mean validation AP over endpoints at seed 0; undefined single-class metrics are skipped. The selected configuration is frozen in results/best_configs.json before final test scoring. The same selected configuration is reused on the random split, which is not separately tuned. Final runs use seeds 0, 1, and 2 for both splits, giving 3 × 2 × 2 × 3 × 12 = 432 endpoint evaluations. Metrics are computed on labeled test rows only.

Table I lists the selected configurations from the repository. The LR and RF grids choose the same configuration for both feature sets, while XGB selects a deeper and slower hybrid configuration. These are validation choices, not test-set choices.

Model	Features	Selected configuration	Validation AP
LR	Morgan	L2, C = 0.03	0.3475
LR	Morgan + descriptors	L2, C = 0.03	0.3429
RF	Morgan	300 trees; no max depth; leaf 1	0.4000
RF	Morgan + descriptors	300 trees; no max depth; leaf 1	0.4433
XGB	Morgan	depth 6; lr 0.03; 600 trees	0.3607
XGB	Morgan + descriptors	depth 7; lr 0.05; 500 trees	0.4307

TABLE I. Selected configurations; validation AP averaged across endpoints, scaffold split, seed 0.

VI. RESULTS

A. Aggregate held-out performance

Table II reports per-split endpoint macro means and sample standard deviations across the three seeds. Every one of the 432 result rows has defined ROC AUC and AP values. RF with descriptors is strongest on scaffold ROC AUC (0.776 ± 0.002) and AP (0.404 ± 0.004). XGB with descriptors is close on scaffold AP at 0.401 ± 0.005. On the random split, XGB with descriptors has the highest ROC AUC (0.862 ± 0.002) and AP (0.565 ± 0.002), narrowly exceeding RF with descriptors in ROC AUC and more clearly in AP.

Model	Features	Scaffold ROC	Scaffold AP	Random ROC	Random AP
LR	Morgan	0.716 ± 0.000	0.312 ± 0.000	0.821 ± 0.000	0.482 ± 0.000
LR	MFP + desc	0.732 ± 0.000	0.312 ± 0.000	0.840 ± 0.000	0.482 ± 0.000
RF	Morgan	0.729 ± 0.006	0.370 ± 0.002	0.836 ± 0.004	0.504 ± 0.001
RF	MFP + desc	0.776 ± 0.002	0.404 ± 0.004	0.861 ± 0.004	0.546 ± 0.004
XGB	Morgan	0.702 ± 0.003	0.318 ± 0.002	0.807 ± 0.001	0.505 ± 0.002
XGB	MFP + desc	0.761 ± 0.004	0.401 ± 0.005	0.862 ± 0.002	0.565 ± 0.002

TABLE II. Test-set endpoint macro means ± sample standard deviation across seeds 0–2. AP is the repository PR-AUC field.

B. Descriptor contribution

Adding descriptors has little effect on LR average precision (+0.0006 ± 0.0000) but improves both tree models on the scaffold split. RF gains +0.0474 ± 0.0074 ROC AUC and +0.0333 ± 0.0031 AP; XGB gains +0.0586 ± 0.0059 ROC AUC and +0.0829 ± 0.0025 AP. These are paired feature-set differences across the three seeds. A possible explanation is that trees combine thresholds on continuous physicochemical properties with binary motif bits; this mechanism was not separately tested.

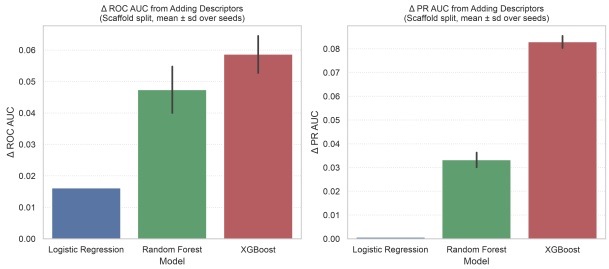


Fig. 7. Scaffold-split performance gain from adding descriptors; bars and error bars show mean and sample SD over three seeds.

C. Scaffold penalty and chemical novelty

For Morgan plus descriptors, the random-minus-scaffold mean gaps are 0.1088 ROC AUC and 0.1695 AP for LR, 0.0845 and 0.1421 for RF, and 0.1015 and 0.1637 for XGB. RF has the smallest hybrid-feature gap. The test sets contain different molecules, so these differences are descriptive and no significance test was run.

Hybrid features	Scaffold ROC	Random ROC	ROC drop	Scaffold AP	Random AP	AP drop
LR	0.7317	0.8405	0.1088	0.3122	0.4818	0.1695
RF	0.7762	0.8607	0.0845	0.4035	0.5456	0.1421
XGB	0.7606	0.8621	0.1015	0.4010	0.5647	0.1637

TABLE III. Random-to-scaffold performance drop for the hybrid representation, computed from the repository test results.

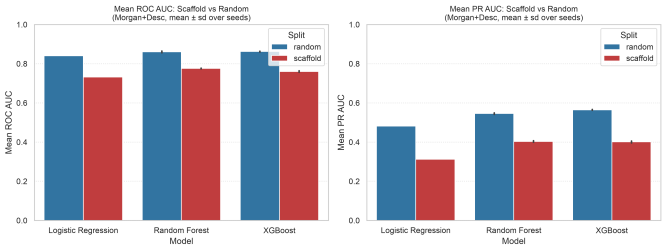


Fig. 8. Hybrid-feature ROC AUC and AP under random and scaffold evaluation.

The supplied Results and Discussion analysis reports maximum Tanimoto similarity from each test molecule to any training molecule using 2,048-bit Morgan fingerprints. It gives random-split mean/median similarity 0.584/0.575 and scaffold-split 0.428/0.396. At threshold 0.85, it reports close training analogues for 72/771 random-test molecules (9.3%) and 13/771 scaffold-test molecules (1.7%). The scaffold test is therefore less similar to training on this analysis, although it is not free of close analogues. These summary values and plot are taken from the supplied analysis PDF; the repository contains the similarity function but not this summary artifact, so we did not independently regenerate them.

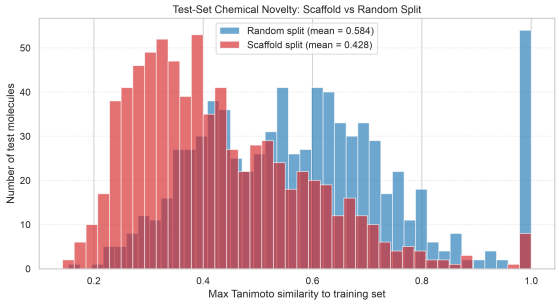


Fig. 9. Test-to-training maximum Tanimoto similarity distributions from the supplied Results and Discussion analysis.

D. Endpoint-level performance

Performance varies by assay. In the hybrid scaffold results, SR-MMP has the highest top model AP (XGB, 0.640 ± 0.004), followed by NR-AhR (RF, 0.596 ± 0.002) and NR-AR-LBD (RF, 0.446 ± 0.003). NR-ER has the lowest best-model ROC AUC (RF, 0.685 ± 0.004). SR-HSE and SR-ATAD5 retain low AP (best 0.267 and 0.247), despite ROC AUC around 0.74–0.78. A plausible explanation for SR-MMP predictability from lipophilicity-related descriptors remains a hypothesis; no causal analysis was conducted.

Endpoint	+ test	LR ROC	RF ROC	XGB ROC	Best AP / model
NR-AR	31	0.691	0.764	0.723	0.464±0.009 XGB
NR-AR-LBD	20	0.775	0.838	0.857	0.446±0.003 RF
NR-AhR	91	0.806	0.841	0.833	0.596±0.002 RF
NR-Aromatase	42	0.695	0.751	0.728	0.367±0.005 RF
NR-ER	79	0.631	0.685	0.674	0.407±0.005 RF
NR-ER-LBD	28	0.705	0.742	0.664	0.310±0.006 RF
NR-PPAR-gamma	30	0.708	0.753	0.739	0.339±0.012 RF
SR-ARE	109	0.702	0.764	0.742	0.526±0.013 RF
SR-ATAD5	33	0.776	0.777	0.757	0.247±0.004 XGB
SR-HSE	41	0.726	0.739	0.727	0.267±0.007 RF
SR-MMP	106	0.836	0.863	0.879	0.640±0.004 XGB
SR-p53	68	0.730	0.796	0.806	0.361±0.001 XGB

TABLE IV. Scaffold test positives, seed-mean ROC AUC by model, and best seed-mean AP/model by endpoint; all use Morgan plus descriptors.

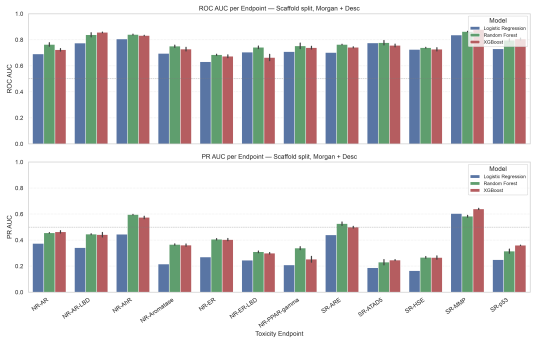


Fig. 10. Endpoint-level scaffold performance for hybrid features; error bars show variation over seeds.

VII. LIMITATIONS AND INTERPRETATION

The main estimates use one fixed scaffold partition and one fixed random partition; three model seeds do not capture split uncertainty. Rare endpoints have few positives, so per-endpoint scores can be noisy. The empty-scaffold convention places every acyclic molecule in training, leaving validation and test restricted to ring-containing structures. Random and scaffold test sets differ in composition, so the score gap is not a paired causal estimate.

The Tanimoto summary and figures are supplied in the Results and Discussion analysis PDF, whereas the checked-in CSV has the model metrics only. Hashed 2,048-bit fingerprints can collide; descriptor correlation can destabilize attribution; and impurity-based feature importance is associative, not causal. The eight-candidate tuning budget is intentionally limited. No significance testing was performed.

VIII. CONCLUSION AND PHASE 2 OUTLOOK

The repository establishes a transparent Phase 1 baseline for 12 masked Tox21 assays. RF with Morgan fingerprints and descriptors is the strongest scaffold macro-average, with hybrid XGB close in AP and strongest on some individual assays. Descriptor augmentation materially benefits the tree models. The supplied novelty analysis supports the interpretation that scaffold-test molecules are less similar to training examples, and the score gap is consistent with the scaffold split being more challenging.

Phase 2 should preserve row order, endpoint masks, splits, and metrics for a fair graph-model comparison. A credible improvement should repeat across endpoints and seeds on scaffold-held-out molecules, with ablations that distinguish the effect of learned graph representations from changes in preprocessing or training.

APPENDIX A. ENDPOINT COVERAGE AND POSITIVE PREVALENCE

Endpoint	Known labels	Actives	Prevalence
NR-AR	7141	304	4.3%
NR-AR-LBD	6652	235	3.5%
NR-AhR	6444	763	11.8%
NR-Aromatase	5718	286	5.0%
NR-ER	6103	781	12.8%
NR-ER-LBD	6851	343	5.0%
NR-PPAR-gamma	6354	185	2.9%
SR-ARE	5739	915	15.9%
SR-ATAD5	6962	261	3.7%
SR-HSE	6374	364	5.7%
SR-MMP	5727	907	15.8%
SR-p53	6673	419	6.3%

TABLE V. Dataset-wide labeled-row counts and active prevalence. PR-AUC chance level is approximately the prevalence shown here.

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